

Topic 3.2 – Disease Dilemmas

“Infectious disease outbreaks, whether natural, intentional or accidental, are still among the foremost dangers to human health and the global economy. With patterns of global travel and trade, disease can spread nearly anywhere within 24 hours.” – Tom Frieden, Director of the Center for Disease Control and Prevention

Diseases do not discriminate who becomes infected or develops symptoms. Diseases can be communicable and noncommunicable and a number of physical and human factors affect an individual's and a community's susceptibility to

the risk. The global nature of some diseases in terms of their geographical spread and scale has encouraged international efforts to combat them.

The causes of disease are often complex and the impacts even more so especially when dealing with these at epidemic and pandemic levels. Continued research into diseases and developments in pharmaceuticals and 'our' understanding of diseases offers opportunities to combat diseases, however unequal access to drugs and information has implications for communities and countries.

1. What are the global patterns of disease and can factors be identified that determine these?	
Key Ideas	Content
1.a. Diseases can be classified and their patterns mapped. The spread of diseases is complex and influenced by a number of factors.	- How diseases can be classified, including infectious and non-infectious, communicable and noncommunicable, contagious and non-contagious, epidemic, endemic and pandemic. - Patterns of diseases, including global distributions of malaria, HIV, tuberculosis, diabetes and cardio-vascular disease. - Disease diffusion and spread to new areas (Hägerstrand model), including the phases of diffusion, physical and socio-economic barriers.
1.b. There is a relationship between physical factors and the prevalence of disease which can change over time.	- Global patterns of temperature, precipitation, relief and water sources and how they affect patterns of disease. - Physical factors can influence vectors of disease such as the prevalence of mosquitoes in warm, humid areas close to water sources. - How seasonal variations influence disease outbreaks such as periods of drought or monsoon rains. - Climate change provides the conditions for emerging infectious diseases to spread to new places and new hosts such as West Nile virus, tsetse fly and tick seasons. - The conditions for zoonotic infectious diseases such as bird flu or rabies to establish and spread from animals to humans.
1.c. Natural hazards can influence the outbreak and spread of disease	Case study of one country which has experienced a natural hazard, such as an earthquake, drought or monsoon rains, and the implications this has on a named disease, such as cholera or typhoid: - geographical area covered by the hazard and its influence on the risk and outbreak of disease - environmental factors affecting the spread of disease such as climate, sanitation, water supply and food - human factors affecting the spread of the disease such as population density, access to clean water, immunisation programmes

2. Is there a link between disease and levels of economic development?

Key Ideas	Content
2.a. As countries develop economically the frequency of communicable diseases decreases, while the prevalence of noncommunicable diseases rises.	• How rising standards of living, including access to food, clean water and sanitation, impact upon susceptibility to disease and influence a country's epidemiological transition. • The reasons why LIDCs have a higher prevalence for communicable diseases (diseases of poverty) and ACs have a higher prevalence for noncommunicable diseases (diseases of affluence). • Case study of one country experiencing air pollution and the impact this has on incidences of cancers (such as lung or bladder). The global and national solutions in dealing with this.

3. How effectively are communicable and noncommunicable diseases dealt with?

Key Ideas	Content
3.a. Communicable diseases have causes and impacts with mitigation and response strategies which have varying levels of success.	• Case study of one communicable disease, such as malaria or tuberculosis, at a country scale, either an LIDC or EDC, including: ◦ environmental and human causes of the disease ◦ prevalence, incidence and patterns of the disease ◦ socio-economic impacts of the disease ◦ direct and indirect strategies used by government and international agencies to mitigate against the disease and respond to outbreaks.
3.b. Noncommunicable diseases have causes and impacts with mitigation and response strategies which have varying levels of success.	• Case study of one noncommunicable disease, such as cardio-vascular disease or diabetes, at a country scale, either an AC or EDC, including: ◦ social, economic and cultural causes of the disease ◦ prevalence, incidence and patterns of the disease ◦ socio-economic impacts of the disease ◦ direct and indirect strategies used by government and international agencies to mitigate against the disease.

4. How far can diseases be predicted and mitigated against?

Key Ideas	Content
4.a. Increasing global mobility impacts the diffusion of disease and the ability to respond to it, at a variety of scales.	• The role of international organisations, such as the World Health Organization, in providing international strategies to combat disease, including predicting diseases, gathering data, research, support programmes and their work with agencies and governments. • Identify a disease outbreak at a global scale, such as H1N1 or SARs, including its rate of spread and patterns of outbreak distribution. • Case study of the role that one NGO has played in dealing with a disease outbreak within one country at national and local level.
4.b. Mitigation strategies to combat global pandemics and overcome physical barriers	• Physical barriers, such as relief, natural hazards, excess water, remoteness of communities, have positive and negative effects on mitigation strategies and response efforts in dealing with diseases. • Mitigation strategies used by government and international agencies to combat global pandemics, such as HIV/AIDS, including screening, availability and funding of treatment and education programmes.

5. Can diseases ever be fully eradicated?

Key Ideas	Content
5.a. Nature has provided medicines to treat disease for thousands of years.	• Medicines from nature, their habitats and conditions for growth including the influence of soil type and climate. • Case study of one medicinal plant, such as rosy periwinkle and opium poppy, including their growing conditions, international trade, medicinal importance for disease and sustainable use. • Conservation issues relating to the international trade in medicinal plants such as endangering species survival, erosion of genetic diversity, threats to the survival of natural ecosystems.
5.b. Top down and bottom up strategies that deal with disease risk and eradication.	• Case study of the global impact of one pharmaceutical transnational, including scientific breakthroughs made, patents, drug manufacturing and their global flows for distribution. • Strategies for disease eradication at a range of scales, including global and national campaigns. • Impact of grassroots strategies in educating communities and the role of women in combating disease risk.